

## CERTIFICATE

This is to certify that the project report entitled “GeoPhysicsLandslideNet: Physics-Closed Pentastream Architecture for Multimodal Landslide Segmentation” submitted by Samreedh Bhuyan, student of Kalinga Institute of Industrial Technology (KIIT), in partial fulfilment of the requirements for the award of the degree of Bachelor of Technology, is a bona fide record of work carried out by him under the guidance of Dr. Anil Earnest at CSIR Fourth Paradigm Institute (CSIR-4PI), Bengaluru. The contents of this report, in full or in parts, have not been submitted to any other Institute or University for the award of any degree or diploma.

Signature of the Guide  
Dr. Anil Earnest  
Scientist / Project Mentor, CSIR-4PI

Signature of the Convenor, SPARK

*A Project Report on*

***GeoPhysicsLandslideNet: Physics-Closed Pentastream  
Architecture for Multimodal Landslide Segmentation***

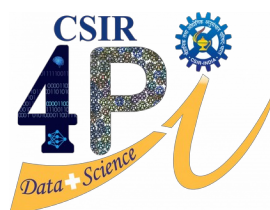
*Submitted in partial fulfilment of requirement for the award of the degree*

**Bachelor of Technology**  
*Of*  
**Kalinga Institute of Industrial Technology (KIIT)**

*By*

Samreedh Bhuyan

*Carried out under the guidance of*  
***CSIR Fourth Paradigm Institute (CSIR-4PI)***



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**Project Report**

**GeoPhysicsLandslideNet: Physics-Closed Pentastream  
Architecture for Multimodal Landslide Segmentation**

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**Bachelor of Technology**

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I gratefully acknowledge the maintainers of the Bije landslide dataset and the Landslide4Sense benchmark for providing open-access remote sensing data used in this study. I also acknowledge the developers of Prithvi-EO, EfficientNet, PyTorch, and the open-source baseline segmentation implementations that enabled fair comparative evaluation.

Finally, I express my heartfelt gratitude to my parents, family members, and friends for their unwavering encouragement, patience, and moral support throughout the completion of this project.

I sincerely thank everyone who has contributed, directly or indirectly, to the successful completion of this work.

Samreedh Bhuyan



# Project Report: GeoPhysicsLandslideNet — Physics-Closed Pentastream Architecture for Multimodal Landslide Segmentation

Samreedh Bhuyan

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Guide: Dr. Anil Earnest  
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## Abstract

This report documents the design, mathematical formulation, implementation, and experimental evaluation of GeoPhysicsLandslideNet (PS-GPLNet), a pentastream physics-closed network for binary landslide segmentation from optical and topographic remote sensing inputs. The architecture embeds Taylor-stabilized infinite-slope factor-of-safety logic throughout encoding, fusion, decoding, and dual-path output equilibrium. Experiments on Bijie and Landslide4Sense demonstrate strong performance against a broad baseline ladder. Implementation lives in `codebase/Geo_physics_equation_derived_model/from_first_steps/`.

## 1 Introduction

Landslide mapping from remote sensing is a central problem in hazard assessment. Purely convolutional segmenters often confuse spectrally similar terrain; dual-stream networks improve matters by separating RGB and DEM pathways, yet their fusion gates remain statistical. Physics-informed geo-AI has advanced susceptibility mapping, but dense segmentation with a closed geotechnical inductive bias spanning the full U-Net-like pipeline is still uncommon.

This report presents PS-GPLNet as an independent architecture: five encoders, Complementary Modality Bridges, MAO-GeoEGCA and TTEB fusion, dual physics decoders with PGDI, and Mechanistic Path Equilibrium Fusion (MPEF). DiGATe-style dual-stream gated U-Nets are used only as baselines.

## 2 Related Work

We position PS-GPLNet against (i) classical and transformer segmentation backbones (U-Net, DeepLabV3+, TransUNet, SegFormer), (ii) dual-stream gated multimodal landslide networks, and (iii) physics-informed susceptibility models that inject Newmark/FS constraints into losses rather than into dense decode routing. The distinguishing claim is architectural closure of FS cells from inlet to outlet.

## 3 Methodology

### 3.1 Factor-of-safety cell

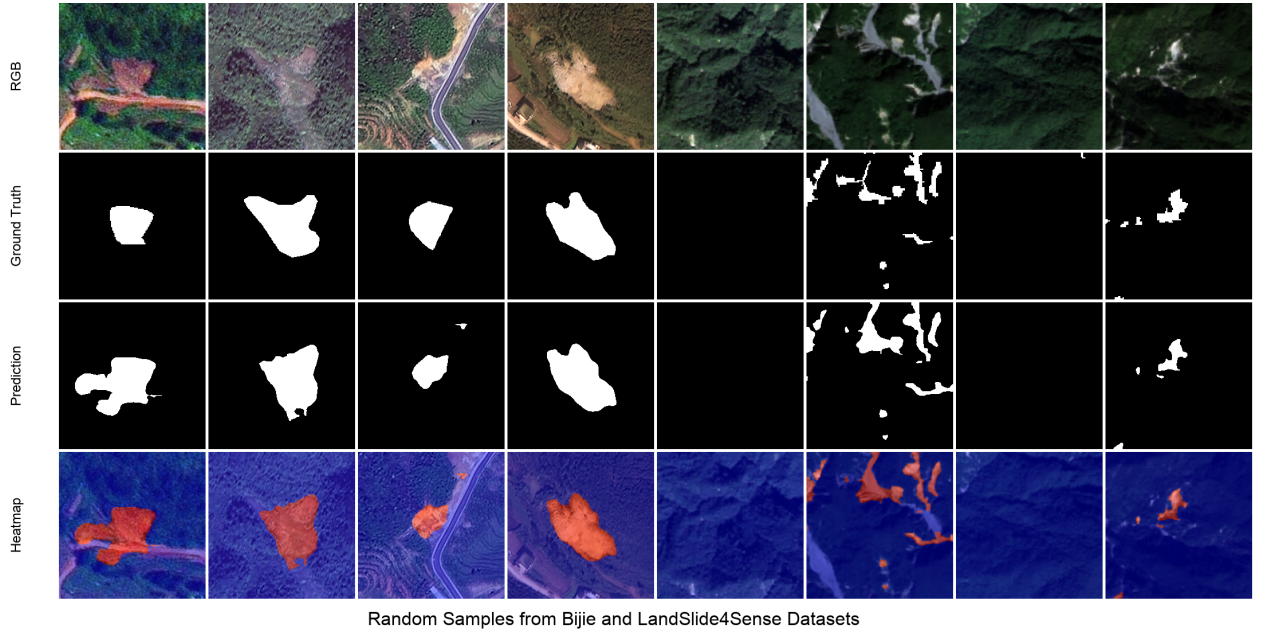
$$FS = \frac{c + \phi h \cos^2 \alpha}{\gamma h \sin \alpha \cos \alpha + w_m m + \varepsilon}, \quad FE = \text{ReLU}(1 - FS).$$

Positive parameters use bounded exponentials for stability during training.

### 3.2 System pipeline

RGB and DEM EfficientNets provide appearance/topography features; physics encoders apply FS-gated cells; Prithvi-EO+LoRA supplies foundation context. CMB hybridizes CNN and physics per modality. MAO-GeoEGCA and TTEB fuse streams. Dual PhysicsDecoders upsample with PGDI; MPEF routes path logits by relative failure energy. Training uses deeply supervised Tversky loss plus MPEF entropy regularization.

A full algorithmic and mathematical description is provided in `from_first_steps/model_architecture.md`.



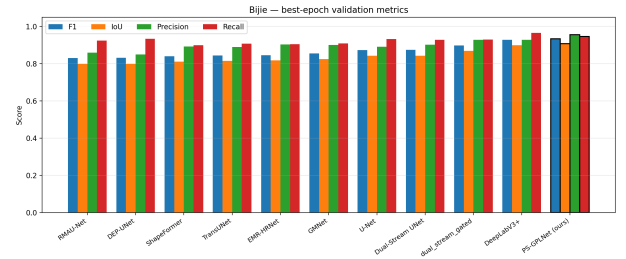
**Figure 1:** Opening qualitative panel (RGB, ground truth, prediction, heatmap) illustrating the landslide segmentation setting.

**Table 1:** Landslide4Sense best-epoch validation (selected models).

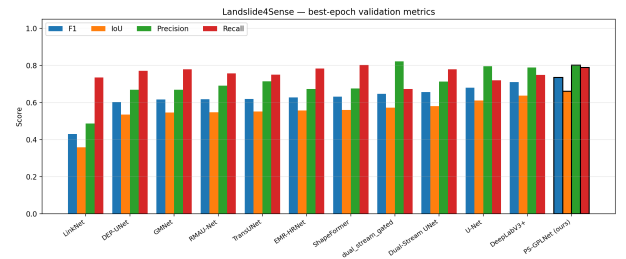
| Model            | F1           | IoU          | Prec  |
|------------------|--------------|--------------|-------|
| LinkNet          | 0.431        | 0.359        | 0.487 |
| Dual-Stream UNet | 0.656        | 0.581        | 0.713 |
| DeepLabV3+       | 0.710        | 0.636        | 0.789 |
| <b>PS-GPLNet</b> | <b>0.736</b> | <b>0.660</b> | 0.802 |

**Table 2:** Bijie best-epoch validation (selected models).

| Model             | F1           | IoU          | Prec         |
|-------------------|--------------|--------------|--------------|
| dual_stream_gated | 0.897        | 0.869        | 0.928        |
| DeepLabV3+        | 0.927        | 0.899        | 0.929        |
| <b>PS-GPLNet</b>  | <b>0.933</b> | <b>0.907</b> | <b>0.956</b> |



**Figure 2:** Bijie comparative metrics including PS-GPLNet.



**Figure 3:** Landslide4Sense comparative metrics including PS-GPLNet.

## 4 Experiments and Results

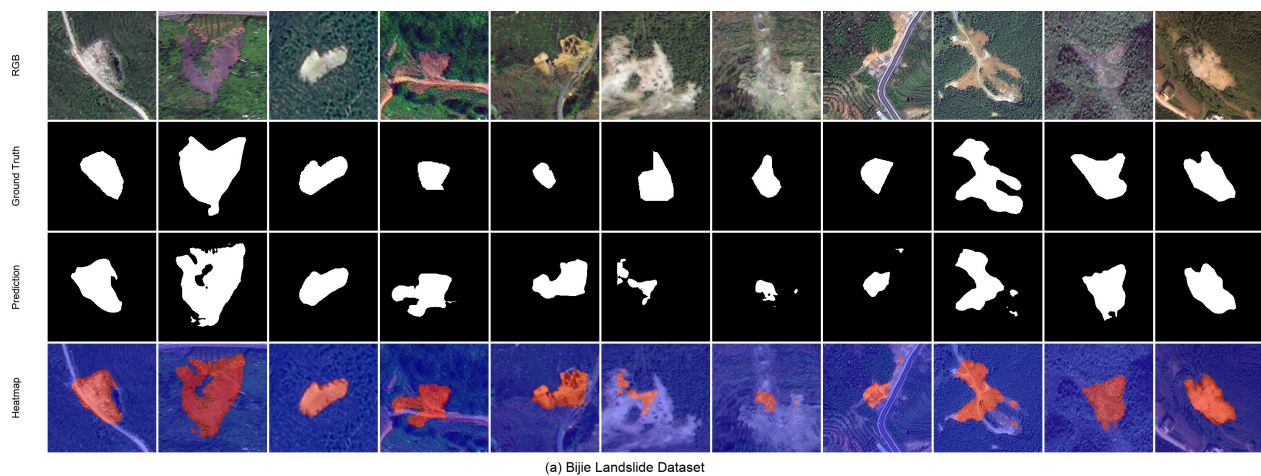
Tables 1–2 summarize best-epoch validation metrics. PS-GPLNet achieves  $F1 = 0.933$  /  $IoU = 0.907$  on Bijie and  $F1 = 0.736$  /  $IoU = 0.660$  on Landslide4Sense.

## 5 Conclusion

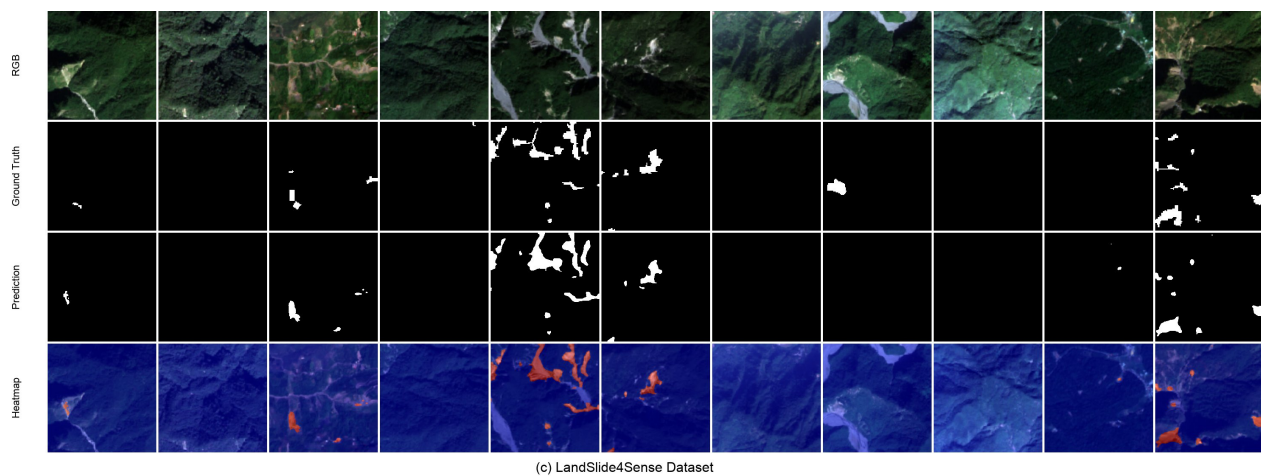
PS-GPLNet demonstrates that a closed geotechnical inductive bias can organize a competitive multimodal landslide segmenter. The report’s companion research paper expands the related-work analysis, algorithms, and full baseline tables for conference submission.

## Acknowledgements

I thank Dr. Anil Earnest and CSIR-4PI for mentorship and facilities, KIIT for academic support, and open dataset/model providers for enabling reproducible comparison.



**Figure 4:** Qualitative Bijie predictions.



**Figure 5:** Qualitative Landslide4Sense predictions.